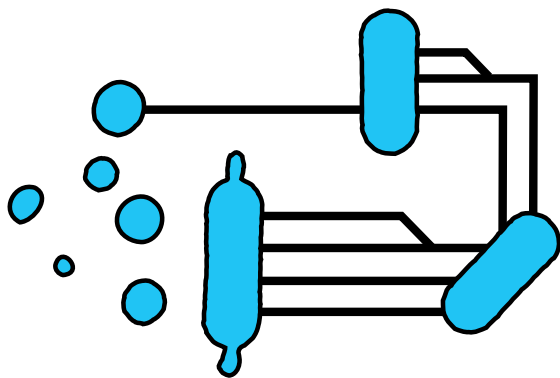




15 - 19 September Belgrade, Serbia

BiomeFUN 2025

From Genomic Analysis to Functional Models
in Microbiomes and Synthetic Consortia



WORKSHOP PROGRAMME

WORKSHOP

**From Genomic Analysis to Functional Models
in Microbiomes and Synthetic Consortia**

BIOMEFUN 2025

15-19 September 2025 Belgrade, Serbia

Scientific & Organizing Committee Chairperson

Ivica Dimkić

University of Belgrade - Faculty of Biology

Scientific & Organizing Committee Co-Chairperson

Đorđe Bajić

Delft University of Technology, the Netherlands

ORGANIZED AND HOSTED BY



University of Belgrade,
Faculty of Biology, Serbia



Delft University of Technology,
the Netherlands

SUPPORTED BY

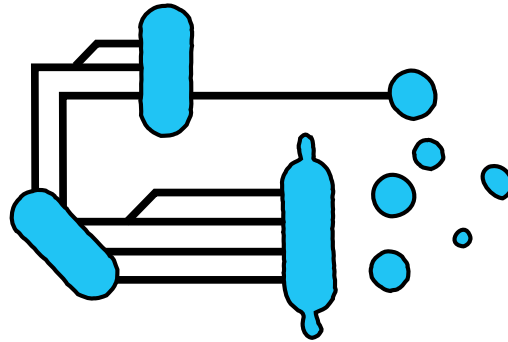


Federation of European
Microbiological Societies (FEMS)



Republic of Serbia
MINISTRY OF SCIENCE,
TECHNOLOGICAL DEVELOPMENT AND INNOVATION

Ministry of Science, Technological
Development and Innovation



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DAY 1: FUNDAMENTALS IN BIOINFORMATICS AND DATA PROCESSING

MONDAY, SEP 15

Chairpersons: *Marcel van den Broek & Milan Dragičević*

8:00 – 9:00 Registration of participants

🕒 9:00 – 9:30 **Welcome and Orientation**
Ivica Dimkić & Djordje Bajić
Introduction by organizers and workshop objectives
Branko Jovčić
FEMS introduction

🕒 9:30 – 10:15 **INVITED LECTURE: Introduction on Sequence Technologies, Common Bioinformatic File Formats and BASH for Genomics**
Marcel van den Broek

10:15 – 10:30 Preparation break for the practical session

🕒 10:30 – 11:30 **Practical 1.1 (part I): Intro Command Line, Cloud, Intro Sequencing, Quality Control, and Filtering**
Marcel van den Broek, Katarina Kruščić

11:30 – 12:00 Coffee break

🕒 12:00 – 13:00 **Practical 1.1 (part II): Intro Command Line, Cloud, Intro Sequencing, Quality Control, and Filtering**
Marcel van den Broek, Katarina Kruščić

13:00 – 14:00 Lunch

🕒 14:00 – 17:00 **Practical 1.2: Amplicon Data Analysis**
Milan Dragičević, Alfonso Esposito

18:00 – 23:00 Poster Party with Catering and Drinks
Opportunity for participants to present their research and network

DAY 2: GENOME AND METAGENOME ANALYSIS

TUESDAY, SEP 16

Chairpersons: *João Pedro Saraiva & Alfonso Esposito*

🕒 9:00 – 09:45 **INVITED LECTURE: Metagenomic Sequencing Approaches for Microbial Communities**
João Pedro Saraiva

09:45 – 10:00 Preparation break for the practical session

🕒 10:00 – 11:30 **Practical 1.3 (part I): Metagenomics Assembly**
Alfonso Esposito, João Pedro Saraiva

11:30 – 12:00 Coffee break

🕒 12:00 – 13:00 **Practical 1.3 (part II): Metagenomics Assembly**
Alfonso Esposito, João Pedro Saraiva

13:00 – 14:00 Lunch

🕒 14:00 – 14:45 **INVITED LECTURE: Functional Annotation of Metagenomes: From Contigs to Community Function**
Prateek Shetty

14:45 – 15:00 Preparation break for the practical session

🕒 15:00 – 18:00 **Practical 1.4: Functional Genomics and Data Visualization**
Prateek Shetty, Bogdan Jovanović

18:10 – 18:20 Group photo in front of the monument to Vojvoda Vuk on Topličin venac square

DAY 3: FUNDAMENTALS OF MODELING: FROM THEORY TO DATA

WEDNESDAY, SEP 17

Chairpersons: *Alexandre Jousset & Djordje Bajić*

🕒 9:00 – 10:00 **KEYNOTE LECTURE: Scalable Digital Twin Framework for Microbiome Engineering and Functional Prediction**
Alexandre Jousset

- 🕒 10:00 – 10:45 **INVITED LECTURE: Modelling Microbial Communities Across Scales to Understand Complexity and Guide Engineering**
Timothy Paez-Watson
- 10:45 – 11:00 Preparation break for the practical session
- 🕒 11:00 – 11:45 **Practical 2.1 (part I): Introduction to Theoretical Models in Microbial Ecology and Python Practice**
Timothy Paez-Watson, Djordje Bajić
- 11:45 – 12:00 Coffee break
- 🕒 12:00 – 13:00 **Practical 2.1 (part II): Introduction to Theoretical Models in Microbial Ecology and Python Practice**
Timothy Paez-Watson, Djordje Bajić
- 13:00 – 14:00 Lunch
- 🕒 14:00 – 14:45 **INVITED LECTURE: From Kinetics to Stoichiometric Models in Metabolic Networks**
Karel Olavarria
- 14:45 – 15:00 Preparation break for the practical session
- 🕒 15:00 – 18:00 **Practical 2.2: Practical Introduction to Metabolic Network Modeling**
Timothy Paez-Watson, Minke Gabriëls

DAY 4: GENOME AND METAGENOME-SCALE METABOLIC MODELING

THURSDAY, SEP 18

Chairpersons: *Karel Olavarria & Prateek Shetty*

- 🕒 9:00 – 09:45 **INVITED LECTURE: Thermodynamic and Kinetic Analysis of Metabolic Networks**
Karel Olavarria
- 09:45 – 10:00 Preparation break for the practical session
- 🕒 10:00 – 11:30 **Practical 2.3: Thermodynamic Analysis of Metabolic Networks**
Karel Olavarria, Minke Gabriëls
- 11:30 – 12:00 Coffee break
- 🕒 12:00 – 13:00 **INVITED LECTURE: Resource Allocation and Dynamic Metabolic Modeling of Microbial Communities**
Djordje Bajić
- 13:00 – 14:00 Lunch
- 🕒 14:00 – 14:45 **Practical 2.4 (part I): Resource Allocation and Dynamic Metabolic Modeling of Microbial Communities**
Djordje Bajić, Uroš Gojković

14:45 – 15:00 Preparation break for the practical session

🕒 **15:00 – 18:00** **Practical 2.4 (part II): Resource Allocation and Dynamic Metabolic Modeling of Microbial Communities**
Djordje Bajić, Uroš Gojković

DAY 5: STATISTICAL APPROACHES AND AI IN MICROBIAL ECOLOGY

FRIDAY, SEP 19

Chairpersons: *Juan Díaz-Colunga & Sascha Patz*

🕒 **9:00 – 09:45** **INVITED LECTURE: Structure-Function Mapping for Microbial Consortia**
Juan Díaz-Colunga

09:45 – 10:00 Preparation break for the practical session

🕒 **10:00 – 11:30** **Practical 3.1 (part I): Structure-Function Landscape Analysis and Prediction**
Juan Díaz-Colunga, Minke Gabriëls

11:30 - 12:00 Coffee break

🕒 **12:00 – 13:00** **Practical 3.1 (part II): Structure-Function Landscape Analysis and Prediction**
Juan Díaz-Colunga, Minke Gabriëls, Uroš Gojković

13:00 – 14:00 Lunch

🕒 **14:00 – 15:00** **INVITED LECTURE: Introduction to Machine Learning in Microbiome Science**
Sascha Patz

🕒 **15:15 – 15:45** **Closing Discussion**
Course Board & Participants
Reflection on workshop outcomes, future directions in microbiome research, and opportunities for collaboration

16:00 – 18:00 **Free Walking Tour: Kalemegdan Park and Belgrade Fortress**

19:00 – 00:00 Unforgettable Gala Dinner
Join us for live music, delicious food, Serbian drinks, engaging conversations, and a celebration of outstanding workshop achievements!

CALL FOR PAPERS:**Thematic Issue on Genomic Analysis and Functional Models in Microbiomes and Synthetic Consortia****EDITORS:**

Ivica Dimkić (University of Belgrade - Faculty of Biology)
& **Djordje Bajić** (Delft University of Technology)

FEMS Microbiology Ecology invites submissions to a thematic issue associated with the BiomeFUN2025 Workshop 'From Genomic Analysis to Functional Models in Microbiomes and Synthetic Consortia', taking place September 15-19, 2025, in Belgrade, Serbia.

Website for more information: <https://bgmicrobiomes.github.io/>

This thematic issue will explore microbial communities related to the health and functionality of ecosystems and the development of sustainable alternatives in different economic sectors. The rapidly evolving field of microbial community ecology and biotechnology requires advanced computational and modeling approaches. To address the growing interest in microbial communities and to utilize the accumulating data — from genome sequencing, environmental studies and metagenomic datasets — advanced computational systems are essential to make sense of this complexity. These systems, which combine bioinformatics with predictive modeling, are at the core of modern efforts to rationally modify microbiomes to achieve specific functional goals. Genomic and metagenomic sequencing are providing unprecedented insights into

the composition and diversity of microbial communities. Recently, new statistical models such as structure-function mapping or various types of machine learning are being developed to predict community behavior and engineer synthetic communities with tailored functional properties.

FEMS Microbiology Ecology welcomes original research articles, mini-reviews, and perspectives that address these topics. All submissions should present complete and novel findings supported by robust experimental data or theoretical insights that contribute to the understanding of microbiomes and synthetic consortia. All manuscripts will undergo standard peer review by members of the Editorial Board and selected experts. Pre-submission inquiries may be directed to the Guest Editors or the Editor in Chief.

SUBMISSION TARGET DATE: 30 December 2025

Authors should specify in their cover letter that their paper is submitted for consideration for the 'Thematic issue on Microbiomes and Synthetic Consortia'. For instructions on submitting a manuscript, please consult FEMS Microbiology Ecology journal instructions at:

https://academic.oup.com/femsec/pages/instructions_for_authors

Accepted manuscripts will be published in regular issues of the journal upon acceptance, and the Thematic Issue will be compiled and made available online upon completion.



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